

Pseudodifferential Operators

1 Part 1: Differential Operators

We would like to generalize the notion of a differential operator on $C^\infty(\mathbb{R}^n)$ to sections of vector bundles on a manifold. Consider the following system of equations:

$$\begin{aligned} 3 \frac{\partial}{\partial y} f(x, y, z) + \frac{\partial}{\partial x} g(x, y, z) + 2 \frac{\partial}{\partial z} h(x, y, z) &= 0 \\ 3 \frac{\partial}{\partial x} f(x, y, z) + 3 \frac{\partial}{\partial x} f(x, y, z) + 3 f(x, y, z) &= 0 \end{aligned}$$

We can represent the right side as an operator D applied to an $\mathbb{R}^3$ valued function, where

$$D = \begin{bmatrix} 3 & 1 & 0 \\ 3 & 1 & 0 \end{bmatrix} \frac{\partial}{\partial x} + \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \frac{\partial}{\partial z} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

In other words, D is a linear map $C^\infty(\mathbb{R}^2, \mathbb{R}^3) \rightarrow C^\infty(\mathbb{R}^2, \mathbb{R}^2)$. This suggests we consider differential operators mapping not just between functions on a manifold, but between sections of bundles E and F .

In the following section, we let X be a manifold, with E and F smooth complex vector bundles on X .

Definition. A *differential operator* of order m on X is a linear map $P : \Gamma(E) \rightarrow \Gamma(F)$, such that if we choose local coordinates x_i in an open set $U \subset X$, and also local trivializations $E|_U \simeq U \times \mathbb{C}^p$ and $F|_U \simeq U \times \mathbb{C}^q$, then P has the form

$$P = \sum_{|\alpha| \leq m} A^\alpha(x) \frac{\partial^{|\alpha|}}{\partial x^\alpha}$$

where the A^α 's are smooth $p \times q$ matrices of smooth functions, and such that one of the highest order coefficients is nonzero somewhere.

Observe that if we make a change of local trivializations of $E|_U$ and $F|_U$ by maps $g_E : U \rightarrow GL_p(\mathbb{C})$ and $g_F : U \rightarrow GL_q(\mathbb{C})$, then in these trivializations, P has the form

$$P = g_F \left(\sum_{|\alpha| \leq m} A^\alpha \frac{\partial^{|\alpha|}}{\partial x^\alpha} \right) g_E^{-1} = \sum_{|\alpha| \leq m} \tilde{A}^\alpha \frac{\partial^{|\alpha|}}{\partial x^\alpha},$$

where the coefficients $\tilde{A}^\alpha$ can be obtained using the product rule to interchange the orders of ∂_x and g_E^{-1} . Namely, for all $|\alpha| = m$, we have the formula

$$\tilde{A}^\alpha = g_E A^\alpha g_E^{-1}$$

for the modified coefficients.

Similarly, if we change coordinates on the set U , P takes the form

$$P = \sum_{|\alpha| \leq m} \tilde{A}^\alpha(\tilde{x}) \frac{\partial^{|\alpha|}}{\partial x^\alpha},$$

and for $|\alpha| = m$, we have

$$\tilde{A}^\alpha = \sum$$